Course Code	Name of the Course					
216U03C603	Optical Fiber Communication					

Teaching Scheme	TH	P		TUT		Total
	03	--		--		03
Credits Assigned	03	--		--		03
Evaluation Scheme	Marks					
	LAB/TUT CA	CA (TH)		ESE		Total
		IA	ISE			
	--	20	30	50		100

Course pre-requisites:
Solid State Physics, Optics, Wave Theory

Course Objectives:
The objective of this course is to understand the fundamental principles of optical fiber technology, optical components and their interactions, behaviour of basic digital/analog systems, analyse the performance characteristics and applications in modern telecommunication systems.

Course Outcomes (CO):
At the end of successful completion of the course the student will be able to
CO1. Apply the principles of optics to optical fiber wave propagation and understand fiber Fabrication techniques
CO2. Understand the transmission characteristics and signal degradation in optical fiber Communication.
CO3. Analyse optical sources and detectors, Power launching and coupling efficiencies.
CO4. Analyse the working of optical Receiver, Digital and Analog systems transmission systems.
CO5. Analyze the optical fiber systems and components required for optical communication Link.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	**Optical Fibers: Structures, Waveguiding and Fabrication**		**07**	**CO1**
	1.1	Optical fiber Structures, Wave guiding and Fabrication: Optical Spectral bands, Basic optical laws and definitions.		
	1.2	Optical fibers Modes and configurations, Single mode fibers, Multi-mode fiber Graded index fiber structures,		
	1.3	Mode theory for circular waveguide, LP modes and power flow.		
	1.4	Fiber materials, fiber fabrication, Mechanical properties.		
2	**Signal Degradation in Optical Fibers**		**12**	**CO 2**
	2.1	Signal Degradation in optical fiber: Attenuation, bending losses, Scattering, Absorption, core cladding losses.		
	2.2	Signal distortion: Dispersion in fibers, limitations due to dispersion, Material, waveguide, polarization dispersion, Intermodal dispersion.		
	2.3	Design optimization of single mode fiber: Dispersion shifted and Dispersion flattened fibers, RI optimization, MFD. bandwidth-length product Non-Linear losses in optical fibers		
3	**Optical Sources, Detectors, Power launching and coupling**		**10**	**CO3**
	3.1	Optical Sources and Detectors : Direct and Indirect band gap materials.		
	3.2	Light Emitting (LED‘s): Structures, quantum efficiency and power, Modulation. Laser Diodes: Laser diode rate equations, Structures and radiation patterns, Single mode lasers, efficiency.		
	3.2	Properties of photo diodes, structures, PIN and APD, photo detector noises, Avalanche Multiplication Noise, Response time, comparison of photodiodes.		
	3.3	Power launching, and coupling, fiber to fiber joints and losses, splicing types, connectors and types.		
4	**Optical Receiver, Digital and Analog systems**		**08**	**CO4**
	4.1	Optical Receiver Operation, receiver structures, pre-amplifier types, Noises and performance calculations.		
	4.2	Point to point links, Digital receiver performance, Link power budget, Rise time budget, transmission distance and noise effects.		
	4.3	Coherent detection Optical detection, Relative intensity noise, Multichannel transmission. amplifiers, basic applications and types, semiconductor optical amplifiers, EDFA		

5	**Optical Fiber Systems and components**		**08**	**CO5**
	5.1	Optical amplifiers, basic applications and types, semiconductor optical amplifiers, EDFA, wavelength converters.		
	5.2	Operational Principles of WDM and DWDM, Standards, Optical Isolators and Circulators, Fiber Grating Filters, Optical Add/Drop Multiplexing, Optical switching networks.		
	5.3	Optical system parameter measurements using OTDR, Eye patterns		
		Total	**45**	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Gerd Keiser	*Optical Fiber*	*Mc- Graw Hill Publication, India*	4th Edition, 2011
2	John Senior	*Optical Fiber Communication*	*Prentice Hall of India Publication*	4th Edition, 2014